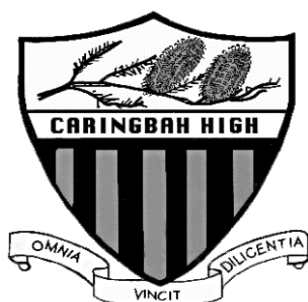


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Student Name



2015 Year 11 Semester 2 Exam

Mathematics Extension 1

General Instructions

- All questions must be attempted
- Start each question on a new page.
- Answers without necessary working may not attract full marks.
- Marks may not be awarded for careless or badly arranged work.
- Diagrams and working which lead to a correct solution may attract some marks.
- Approved calculators may be used.

Time Allowed

1 hour 30 min. + 3 min. reading time

Question 1 (13 marks)

- a) The point P divides the interval joining $A(-1, -2)$ to $B(9, 3)$ internally in the ratio $4:1$. Find the coordinates of P . (2)

- b) Find the domain of the function (2)

$$f(x) = \sqrt{2x+3} + \sqrt{x - \frac{1}{2}}$$

- c) Sketch the piece meal function ($\frac{1}{3}$ page size) (3)

$$f(x) = \begin{cases} x^2 + 1 & x \geq 0 \\ x - 3 & x < 0 \end{cases}$$

- d) The gradient of the tangent to $y = 3x^2 - mx + 1$ has a value of 2 at $x = -1$, find the value of m . (2)

- e) i) Prove that (2)

$$\frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

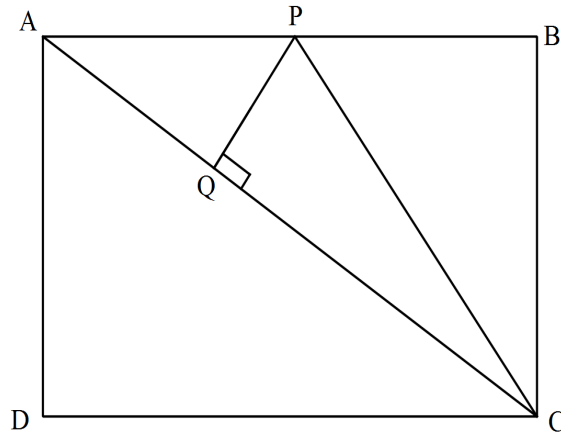
- ii) Use the result above to find $\frac{d}{dx}(\sqrt{x})$ from first principles. (2)

Student Name:

Question 2 (13 marks)

- a) In the diagram $ABCD$ is a square, PC bisects $\angle ACB$ and Q is the foot of the perpendicular from P to AC

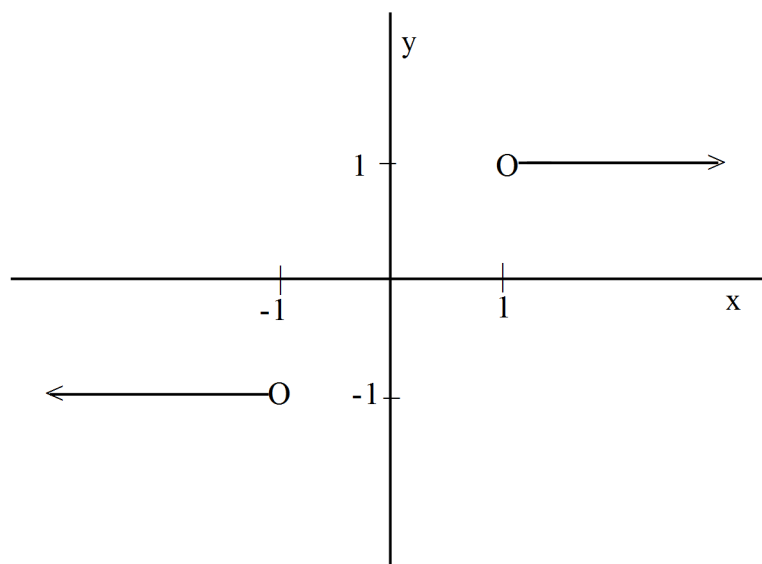
(Diagram not to scale)



Copy the diagram into your booklet.

- i) Prove $\triangle PBC \equiv \triangle PQC$ (3)
- ii) Prove that the area of $\triangle APQ$ is $\frac{x^2}{6} \text{ cm}^2$ given that $BC = x \text{ cm}$ and $PB = \frac{1}{3} AB$ (3)

- b) Sketch a possible function that could have the gradient function as graphed below. ($\frac{1}{3}$ page size) (2)



- c) Find the derivative of the following

i) $y = \frac{1}{3}x^3 - 2x^2 + 6$ (1)

ii) $y = \frac{(4x+3)^3}{x+1}$ (2)

iii) $y = (4x-1)\sqrt{2x+4}$ (2)

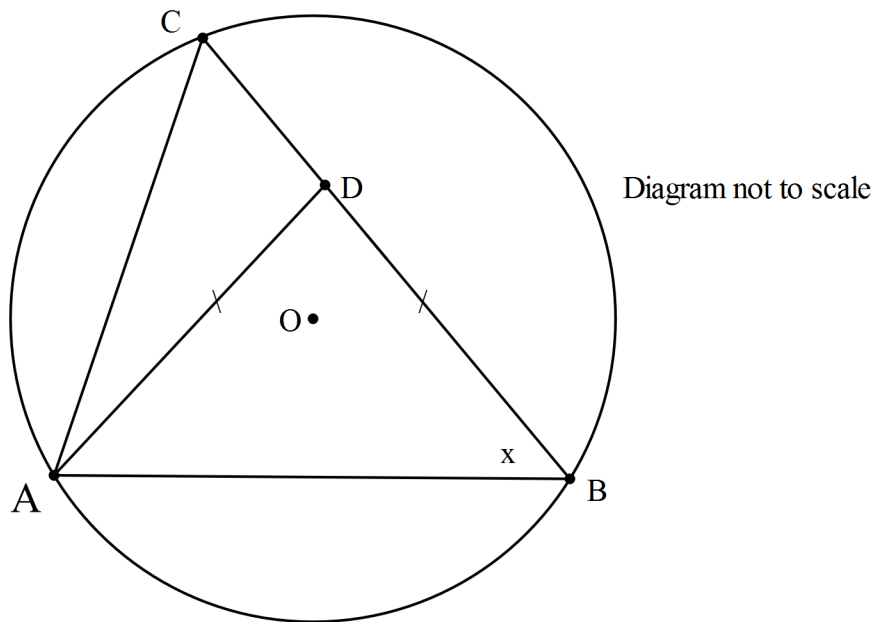
Question 3 (13 marks)

- a) On the same diagram ($\frac{1}{3}$ page size) shade in the region for which (3)

$$\begin{cases} y < x^2 \\ y \geq 0 \\ y \leq \sqrt{9-x^2} \end{cases}$$

are simultaneously true.

- b) In the diagram, the vertices of $\triangle ABC$ lie on the circle with centre O .
The point D lies on BC such that $\triangle ABD$ is isosceles $AD = BD$ and $\angle ABC = x$



Copy or trace this diagram into your writing booklet.

- i) Explain why $\angle AOC = 2x$ (2)
- ii) Prove $ACDO$ is a cyclic quadrilateral. (2)
- c) Solve for x : $4^x - 6(2^x) + 8 = 0$ (2)

- d) The roots of the equation $x + \frac{1}{x} = 5$ are α and β .

Find the value of

i) $\alpha + \frac{1}{\alpha}$ (1)

ii) $\alpha + \beta$ (1)

iii) $\alpha^2 + \beta^2$ (2)

Question 4 (13 marks)

- a) For the function $y = x^2 - 4x$, find :

i) the co ordinates of points of intersection with the x axis. (1)

ii) the gradient of the tangents at these points of intersection. (2)

iii) Find the acute angle formed by the intersection of these two tangents. (2)

- b) Find the values of K for which $x^2 + Kx + (K+1)^2 = 0$ has two roots. (2)

- c) The general form of the line, l , through the intersection of the two lines

$$5x - 2y + 3 = 0 \quad \text{and} \quad 2x + 6y - 7 = 0$$

$$\text{is given by } l : (5x - 2y + 3) + k(2x + 6y - 7) = 0$$

(You do not need to show this)

i) Show the gradient of l is $\frac{5+2k}{2-6k}$ (1)

ii) Hence find the equation of the line, l , if it is parallel to $2y - x + 4 = 0$ (2)
Write your equation in general form.

d) Prove that the line $5x - 12y = 52$ is tangent to the circle with equation (3)

$$x^2 + y^2 = 16.$$

Question 5 (13 marks)

a) Consider the function defined by

$$f(x) = \frac{2x-2}{x^2+x-2}$$

i) For what values of x is $f(x)$ discontinuous? (2)

ii) At each point of discontinuity found in part i) determine whether $f(x)$ has a limit, and if so, give the value of the limit. (2)

iii) Write an equation for each vertical and horizontal asymptote to the graph of $y = f(x)$. Justify your answer. (2)

iv) Draw a sketch of $f(x) = \frac{2x-2}{x^2+x-2}$ showing above features ($\frac{1}{3}$ page size) (3)

b)

i) Find the value of k such that $y = x + k$ is a tangent to $y = x^2 - x - 2$ (2)

ii) What are the co ordinates of the point of contact ? (2)

END OF EXAM

Question 1

(c)

(a) x_1, y_1, x_2, y_2
 $(-1, -2), (9, 3)$

$4 : 1$
 $m : n$

$$x = \frac{1(-1) + 4(9)}{5}$$

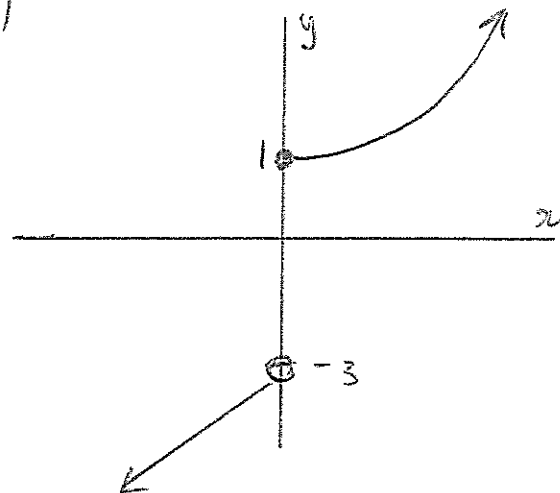
$$= \frac{35}{5}$$

$$= 7$$

$$y = \frac{1(-2) + 4(3)}{5}$$

$$= \frac{10}{5}$$

$$= 2$$



(d) $y = 3x^2 - mx + 1$

$$y' = 6x - m$$

$$2 = 6(-1) - m$$

$$m = 8$$

(e) (i) $\frac{1}{\sqrt{x+h} + \sqrt{x}} \times \frac{\sqrt{x+h} - \sqrt{x}}{\sqrt{x+h} - \sqrt{x}}$

$$= \frac{\sqrt{x+h} - \sqrt{x}}{x+h - x}$$

$$= \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

(ii) $\frac{\sqrt{x+h} - \sqrt{x}}{h}$

$$= \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

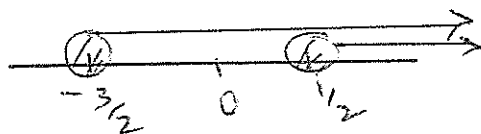
$$\lim_{h \rightarrow 0} = \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

(d) $f(x) = \sqrt{2x+3} + \sqrt{x-\frac{1}{2}}$

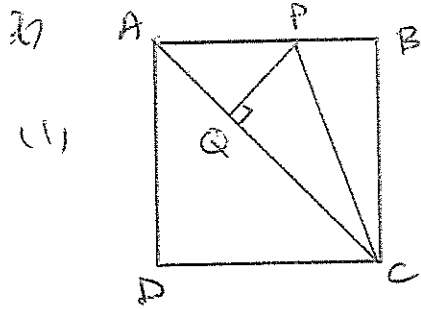
$$2x+3 \geq 0 \quad x - \frac{1}{2} \geq 0$$

$$x \geq -\frac{3}{2} \quad x \geq \frac{1}{2}$$



D: $x \geq \frac{1}{2}$

Question 2



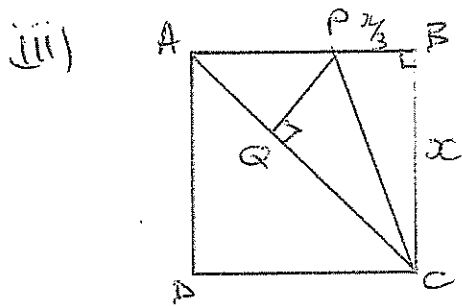
In $\triangle PBC$ and $\triangle PQC$
 $\angle PQC = 90^\circ$ (given $PQ \perp AC$)

(ii) $\angle PBC = 90^\circ$ (prop of square)
 $\therefore \angle PQC = \angle PBC$

$\angle QCP = \angle PCB$ (given)

PC is common side

$\triangle PBC \equiv \triangle PQC$ (A.A.S)



$$\text{Area } \triangle ABC = \frac{1}{2}x^2$$

$$\begin{aligned} \text{Area } \triangle PBC &= \frac{1}{2} \times x \times \frac{2x}{3} \\ &= \frac{x^2}{3} \end{aligned}$$

$$\text{Area } \triangle PBC = \text{Area } \triangle PQC \text{ (cong } \triangle\text{)}$$

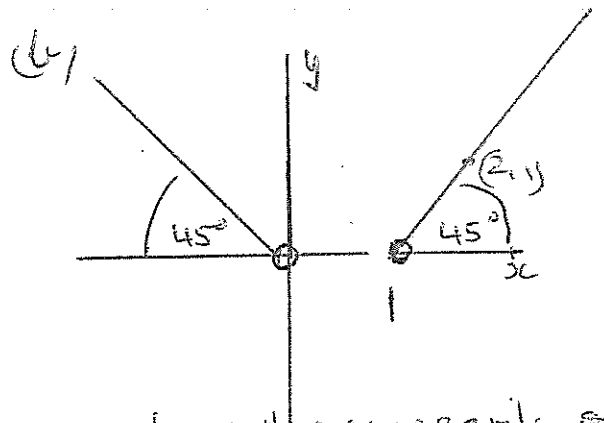
$$\text{Area } \triangle ABC = 2 \text{Area } \triangle PBC$$

$$\frac{1}{2}x^2 = 2 \left[\frac{x^2}{3} \right]$$

$$= \frac{1}{2}x^2 = \frac{x^2}{3}$$

$$= \frac{x^2}{6} \text{ cm}^2$$

$$\frac{3(4x+5)}{\sqrt{2x+4}} \leftarrow$$



Note not necessarily on x axis

(c) $y = \frac{1}{3}x^3 - 2x^2 + 6$

(i) $y' = x^2 - 4x$

(ii) $y = \frac{(4x+3)^3}{x+1}$

$$y' = \frac{(x+1) \cdot 3(4x+3)^2 \cdot 4 - (4x+3)^3 \cdot 1}{(x+1)^2}$$

$$= \frac{(4x+3)^2 [12(x+1) - (4x+3)]}{(x+1)^2}$$

$$= \frac{(4x+3)^2 [8x+9]}{(x+1)^2}$$

(iii) $y = (4x-1)(2x+4)^{1/2}$

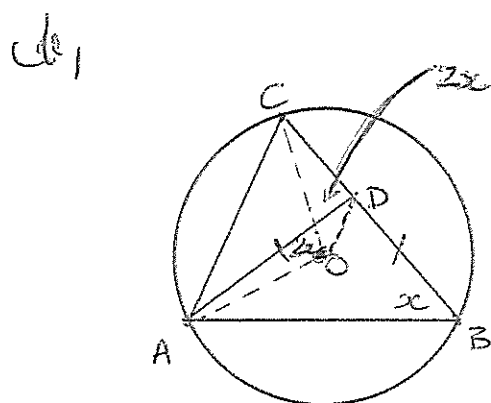
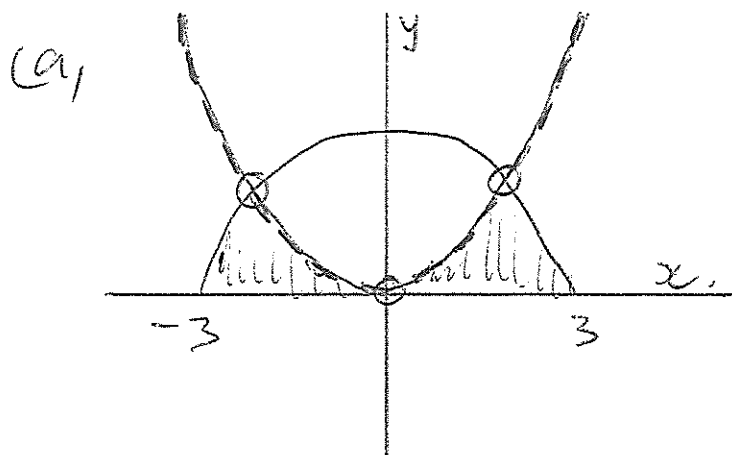
$$y' = (2x+4)^{1/2} \cdot 4 + (4x-1) \cdot \frac{1}{2}(2x+4)^{-1/2}$$

$$= 4(2x+4)^{1/2} + \frac{4x-1}{(2x+4)^{1/2}}$$

$$= \frac{4(2x+4) + 4x-1}{\sqrt{2x+4}}$$

$$= \frac{(12x+15)}{\sqrt{2x+4}}$$

Question 3.



(i) $\angle AOC = 2x$

Angle at the centre twice
the angle at the circ standing
on same chord AC or arc AC

(ii) Join OD.

$\angle OAB = x$ base angles of
isos $\triangle AOB$ equal

$\angle ODA = 2x$ ext $\angle$ of $\triangle AOB$
= to 2 opp int $\angle$'s

$\angle ODA = \angle OCA$
 $= 2x$

CAOD cyclic quad angles at
circ standing on same chord are equal.

(d) $x + \frac{1}{x} = 5$

$$x + \frac{1}{x} - 5 = 0$$

$$x^2 - 5x + 1 = 0$$

(i) 5

(ii) 5

(iii) $x^2 + y^2$

$$= (x+y)^2 - 2xy$$

$$= 5^2 - 2(1)$$

$$= 23$$

(c) $4^x - 6(2^x) + 8 = 0$

$$(2^x)^2 - 6(2^x) + 8 = 0$$

let $m = 2^x$

$$m^2 - 6m + 8 = 0$$

$$(m-4)(m-2) = 0$$

$$m = 4, \quad m = 2$$

$$2^x = 4 \quad 2^x = 2$$

$$x = 2, \quad x = 1$$

Question 4

(a) $y = x^2 - 4x$

when $y = 0$

(i) $x(x - 4) = 0$

$$x = 0, \quad x = 4$$

(ii) $y' = 2x - 4$

at $x = 0$

$$m_T = -4$$

at $x = 4$

$$m_T = 4$$

(ii) $\tan \phi = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$= \left| \frac{4 - (-4)}{1 + 4(-4)} \right|$$

$$\tan \phi = \frac{8}{15}$$

$$\phi = 28^\circ$$

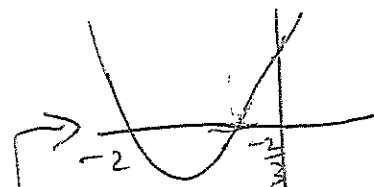
4, $k^2 - 4(1)(k+1)^2 > 0$

$$k^2 - 4(k^2 + 2k + 1) > 0$$

$$k^2 - 4k^2 - 8k - 4 > 0$$

$$\therefore 3k^2 + 8k + 4 < 0$$

$$(3k+2)(k+2) < 0$$



$$-2 < k < -2/3$$

Question 4

$$(C) (5x - 2y + 3) + k(2x + 6y - 7) = 0$$

$$5x - 2y + 3 + 2kx + 6ky - 7k = 0$$

$$(i) -y(2 - 6k) + x(5 + 2k) + 3 - 7k = 0$$

$$y(2 - 6k) = (5 + 2k)x + 3 - 7k$$

$$y = \left(\frac{5 + 2k}{2 - 6k} \right)x + \frac{3 - 7k}{2 - 6k}$$

$$\therefore M = \frac{5 + 2k}{2 - 6k}$$

$$(ii) \quad 2y - x + 4 = 0$$
$$2y = x - 4$$
$$y = \frac{x}{2} - 2$$

$$M = \frac{1}{2}$$

$$\frac{5 + 2k}{2 - 6k} = \frac{1}{2}$$

$$10 + 4k = 2 - 6k$$

$$8 = -10k$$

$$k = -\frac{4}{5}$$

Required eqn

$$5x - 2y + 3 - \frac{4}{5}(2x + 6y - 7) = 0$$

$$25x - 10y + 15 - 8x - 24y + 28 = 0$$

$$17x - 34y + 43 = 0$$

$$(d) \quad 5x - 12y - 52 = 0$$

$$a = 5, b = -12, c = -52$$

$$(0, 0)$$

$$P.D. = \frac{|5(0) + 12(0) - 52|}{\sqrt{5^2 + (-12)^2}}$$

$$= \frac{52}{13}$$

$$= 4$$

now radius of circle is 4 and $\perp$ dist from centre to point of contact is 4

$\therefore 5x - 12y = 52$ tangent to circle.

Question 5

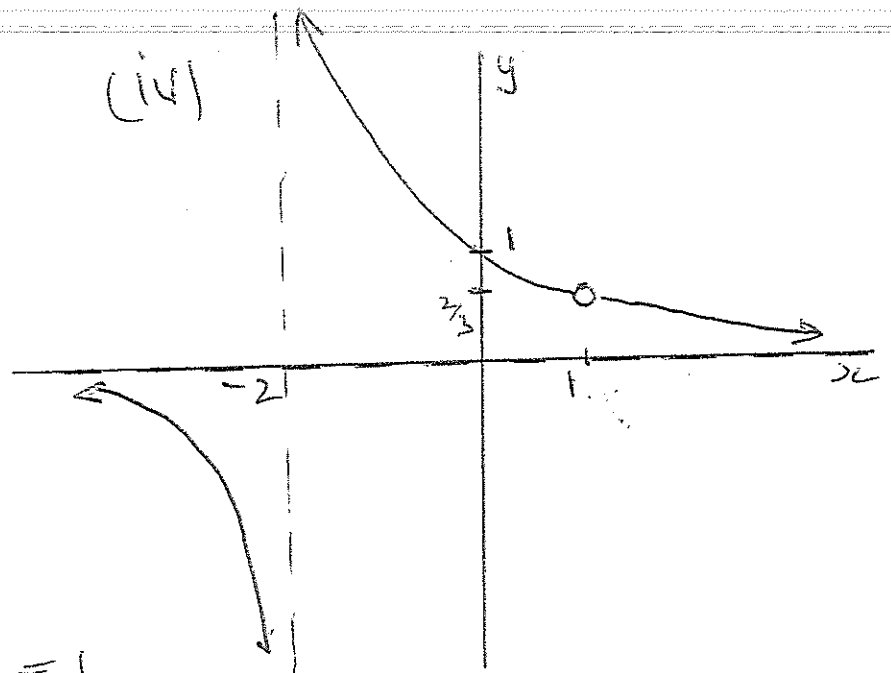
$$f(x) = \frac{2x - 2}{x^2 + x - 2}$$

(a)

$$(i), f(x) = \frac{2x - 2}{(x+2)(x-1)}$$

Discontinuous

at $x = -2, x = 1$



$$(ii) \lim_{x \rightarrow -2} \frac{2(x-1)}{(x+2)(x-1)}$$

$$= \frac{2}{0} \\ \rightarrow \infty$$

$$\lim_{x \rightarrow 1} \frac{2(x-1)}{(x+2)(x-1)} \\ = \frac{2}{3}$$

(iii) Vert asymptote

at $x = -2$

from (ii)

Horizontal

$$\lim_{x \rightarrow \infty} \frac{\frac{2x}{x^2} - \frac{2}{x^2}}{\frac{x^2}{x^2} + \frac{x}{x^2} - \frac{2}{x^2}}$$

$$= 0 \therefore x \text{ axis}$$

$$(i) y = x^2 - x - 2 \\ y = x + k$$

$$(ii) x^2 - x - 2 = x + k \\ x^2 - x - 2 - x - k = 0 \\ x^2 - 2x - 2 - k = 0 \\ \Delta = 0 \text{ for one soln.} \\ \Delta = (-2)^2 - 4(1)(-2-k) \\ 0 = 4 + 8 + 4k \\ 0 = 12 + 4k \\ k = -3$$

$$(ii) x^2 - 2x - 2 + 3 = 0$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1 \quad y = -2$$